

11.2

1) an equation of an ellipse is given.

$$\frac{x^2}{100} + \frac{y^2}{64} = 1$$

Find vertices, foci, and eccentricity.

x-major
 $100 = a^2$
 $a = \sqrt{100} = 10$
 $64 = b^2$
 $b = \sqrt{64} = 8$

vertex: $(-10, 0)$
 vertex: $(10, 0)$

$c^2 = a^2 - b^2$
 $= 100 - 64$
 $c^2 = 36$
 $c = 6$

Focus: $(-6, 0)$
 Focus: $(6, 0)$

$e = \frac{6}{10} = \frac{3}{5}$

eccentricity: $\frac{3}{5}$

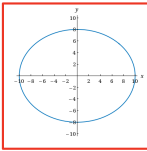
length of major axis

$2a = 2 \cdot 10 = 20$

length of minor axis

$2b = 2 \cdot 8 = 16$

graph



2) an equation of an ellipse is given.

$$2x^2 + 64y^2 = 128$$

Find vertices, foci, and eccentricity.

$$\frac{x^2}{64} + \frac{y^2}{2} = 1$$

x-major
 $64 = a^2$
 $a = 8$
 $2 = b^2$
 $b = \sqrt{2}$

vertex: $(-8, 0)$
 vertex: $(8, 0)$

$c^2 = a^2 - b^2$
 $= 64 - 2$
 $c^2 = 62$
 $c = \sqrt{62}$

Focus: $(-\sqrt{62}, 0)$
 Focus: $(\sqrt{62}, 0)$

$e = \frac{\sqrt{62}}{8}$

eccentricity: $\frac{\sqrt{62}}{8}$

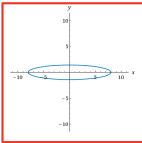
length of major axis

$2a = 2 \cdot 8 = 16$

length of minor axis

$2b = 2\sqrt{2} = 2\sqrt{2}$

graph



3) an equation of an ellipse is given.

$$y^2 = 1 - 3x^2$$

Find vertices, foci, and eccentricity.

$$y^2 + 3x^2 = 1$$

$$\frac{x^2}{\frac{1}{3}} + \frac{y^2}{1} = 1$$

y-major

$$1 = a^2$$

$$a = 1$$

$$\frac{1}{3} = b^2$$

$$b = \frac{\sqrt{3}}{2}$$

$$\text{vertex: } (0, -1)$$

$$\text{vertex: } (0, 1)$$

$$c^2 = a^2 - b^2$$

$$= 1 - \frac{1}{3}$$

$$c^2 = \frac{2}{3}$$

$$c = \frac{\sqrt{2}}{\sqrt{3}}$$

$$= \frac{\sqrt{6}}{3}$$

$$\text{Focus: } (0, -\frac{\sqrt{6}}{3})$$

$$\text{Focus: } (0, \frac{\sqrt{6}}{3})$$

$$e = \frac{\frac{\sqrt{6}}{3}}{1} = \frac{\sqrt{6}}{3}$$

$$\text{eccentricity: } \frac{\sqrt{6}}{3}$$

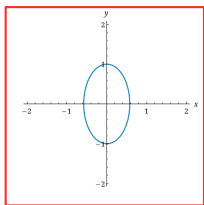
length of major axis

$$2a = 2 \cdot 1 = 2$$

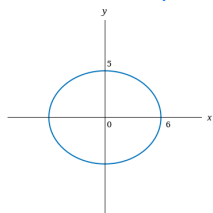
length of minor axis

$$2b = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$$

graph



4) Find standard equation for ellipse whose graph is shown



x-major

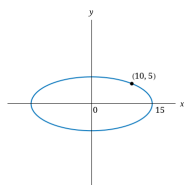
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$a = 6$$

$$b = 5$$

$$\frac{x^2}{36} + \frac{y^2}{25} = 1$$

5) Find standard equation for ellipse whose graph is shown



x-major

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$a = 15$$

$$\frac{10^2}{225} + \frac{25}{b^2} = 1$$

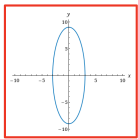
$$\frac{125}{b^2} = \frac{125}{225}$$

$$b^2 = 45$$

$$\frac{x^2}{225} + \frac{y^2}{45} = 1$$

6) Use a graphing device to graph

$$9x^2 + y^2 = 81$$



7) Find standard equation for ellipse that satisfies given conditions

Foci: $(0, \pm 4)$ vertices: $(0, \pm 5)$

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$a = 5$$

$$c = 4$$

$$c^2 = a^2 - b^2$$

$$16 = 25 - b^2$$

$$b^2 = 9$$

$$\frac{x^2}{9} + \frac{y^2}{25} = 1$$

8) Find standard equation for ellipse that satisfies given conditions

Foci: $F(\pm\sqrt{15}, 0)$, vertices: $(\pm 8, 0)$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$c = \sqrt{15}$$

$$a = 8$$

$$c^2 = a^2 - b^2$$

$$15 = 64 - b^2$$

$$b^2 = 49$$

$$\frac{x^2}{64} + \frac{y^2}{49} = 1$$

9) Find standard equation for ellipse that satisfies given conditions

Length of major axis: 8, length of minor axis: 6, foci on y-axis, centered at origin

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$2a = 8$$

$$a = 4$$

$$2b = 6$$

$$b = 3$$

$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

10) Find standard equation for ellipse that satisfies given conditions

Endpoints of major axis: $(\pm 9, 0)$, distance between foci: 4

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$a = 9$$

$$c = 2$$

$$c^2 = a^2 - b^2$$

$$4 = 81 - b^2$$

$$b^2 = 77$$

$$\frac{x^2}{81} + \frac{y^2}{77} = 1$$

11) Find standard equation for ellipse that satisfies given conditions

Eccentricity: 0.25, foci: $(\pm 0.75, 0)$

x-major

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$c = \frac{3}{4}$$

$$\frac{\frac{3}{4}}{a} = \frac{1}{4}$$

$$a = 3$$

$$c^2 = a^2 - b^2$$

$$\frac{9}{16} = 9 - b^2$$

$$b^2 = \frac{135}{16}$$

$$\boxed{\frac{x^2}{9} + \frac{y^2}{\frac{135}{16}} = 1}$$

12) Find intersection pts of pair of ellipses

$$\begin{cases} 200x^2 + 50y^2 = 200 \\ x^2 + \frac{y^2}{25} = 1 \end{cases}$$

$$\begin{cases} x^2 + \frac{y^2}{4} = 1 \\ x^2 + \frac{y^2}{25} = 1 \end{cases}$$

$$\left(\left(x^2 + \frac{y^2}{4} \right) - \left(x^2 + \frac{y^2}{25} \right) \right) = 1 - 1$$

$$\frac{y^2}{4} - \frac{y^2}{25} = 0$$

$$\frac{25y^2}{100} - \frac{4y^2}{100} = 0$$

$$\frac{21y^2}{100} = 0$$

$$21y^2 = 0$$

$$y^2 = 0$$

$$y = 0$$

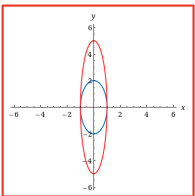
$$x^2 + \frac{0^2}{4} = 1$$

$$x^2 = 1$$

$$x = \pm 1$$

$$\boxed{\begin{matrix} (-1, 0) \\ (1, 0) \end{matrix}}$$

Graph



13) An equation of a hyperbola is given.

$$\frac{y^2}{25} - \frac{x^2}{4} = 1$$

Find vertices, foci, and asymptotes.

y-major

$$a = \sqrt{25}$$

$$a = 5$$

$$\text{vertex: } (0, -5)$$

$$\text{vertex: } (0, 5)$$

$$b^2 = 4$$

$$c^2 = a^2 + b^2$$

$$c^2 = 25 + 4$$

$$c = \sqrt{29}$$

$$\text{Foci: } (0, -\sqrt{29})$$

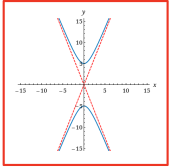
$$\text{Foci: } (0, \sqrt{29})$$

$$\text{asymptotes: } y = \pm \frac{5}{2}x$$

length of transverse axis

$$2a = 2 \cdot 5 = 10$$

graph



14) An equation of a hyperbola is given.

$$\frac{x^2}{6} - y^2 = 1$$

Find vertices, foci, and asymptotes.

x-major

$$a^2 = 6$$

$$a = \sqrt{6}$$

$$\text{Vertex: } (-\sqrt{6}, 0)$$

$$\text{Vertex: } (\sqrt{6}, 0)$$

$$b^2 = 1$$

$$b = 1$$

$$c^2 = a^2 + b^2$$

$$c^2 = 6 + 1$$

$$c = \sqrt{7}$$

$$\text{Foci: } (-\sqrt{7}, 0)$$

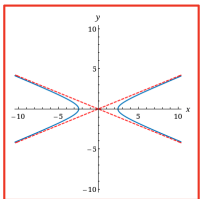
$$\text{Foci: } (\sqrt{7}, 0)$$

$$\text{asymptotes: } y = \pm \frac{1}{\sqrt{6}} = \pm \frac{\sqrt{6}}{6}x \rightarrow y = \pm \frac{\sqrt{6}}{6}x$$

length of transverse axis

$$2 \cdot a = 2 \cdot \sqrt{6} = 2\sqrt{6}$$

Graph



15) An equation of a hyperbola is given.

$$25x^2 - 16y^2 = 400$$

Find vertices, foci, and asymptotes.

$$\frac{x^2}{16} - \frac{y^2}{25} = 1$$

x-major

$$a^2 = 16$$

$$a = 4$$

$$\text{vertex: } (-4, 0)$$

$$\text{vertex: } (4, 0)$$

$$b^2 = 25$$

$$b = 5$$

$$c^2 = a^2 + b^2$$

$$c^2 = 16 + 25$$

$$c = \sqrt{41}$$

$$\text{Foci: } (-\sqrt{41}, 0)$$

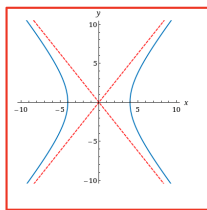
$$\text{Foci: } (\sqrt{41}, 0)$$

$$\text{asymptotes: } y = \pm \frac{5}{4}x$$

length of transverse axis

$$2a = 2 \cdot 4 = 8$$

graph



16) An equation of a hyperbola is given.

$$9y^2 - 4x^2 = 36$$

Find vertices, foci, and asymptotes.

$$\frac{y^2}{4} - \frac{x^2}{9} = 36$$

y-major

$$a^2 = 4$$

$$a = 2$$

$$\text{Vertex: } (0, -2)$$

$$\text{Vertex: } (0, 2)$$

$$b^2 = 9$$

$$b = 3$$

$$c^2 = a^2 + b^2$$

$$c^2 = 4 + 9$$

$$c = \sqrt{13}$$

$$\text{Foci: } (0, -\sqrt{13})$$

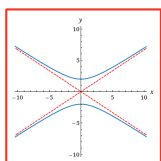
$$\text{Foci: } (0, \sqrt{13})$$

$$\text{Asymptotes: } y = \pm \frac{2}{3}x$$

length of transverse axis

$$2a = 2 \cdot 2 = 4$$

graph



17) An equation of a hyperbola is given.

$$x^2 - 3y^2 + 108 = 0$$

Find vertices, foci, and asymptotes.

$$x^2 - 3y^2 = -108$$

$$-\frac{x^2}{108} + \frac{y^2}{36}$$

$$\frac{y^2}{36} - \frac{x^2}{108}$$

y-major

$$a^2 = 36$$

$$a = 6$$

$$\text{Vertex: } (0, -6)$$

$$\text{Vertex: } (0, 6)$$

$$b^2 = 108$$

$$c^2 = a^2 + b^2$$

$$= 36 + 108$$

$$= 144$$

$$c = 12$$

$$\text{Foci: } (0, -12)$$

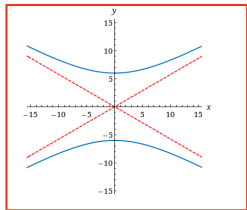
$$\text{Foci: } (0, 12)$$

$$\text{asymptotes: } y = \pm \frac{6}{\sqrt{108}} x = \pm \frac{6\sqrt{108}}{108} x \rightarrow y = \pm \frac{6\sqrt{108}}{108} x$$

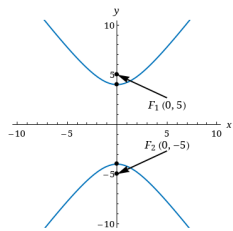
length of transverse axis

$$2a = 2 \cdot 6 = 12$$

graph



18) Find the standard equation for the hyperbola whose graph is shown



y-major

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$a = 4$$

$$c = 5$$

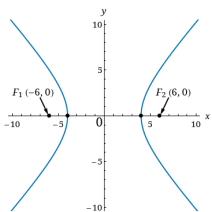
$$c^2 = a^2 + b^2$$

$$25 = 16 + b^2$$

$$b^2 = 9$$

$$\frac{y^2}{16} - \frac{x^2}{9} = 1$$

19) Find the standard equation for the hyperbola whose graph is shown



x-major

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$a = 4$$

$$c = 6$$

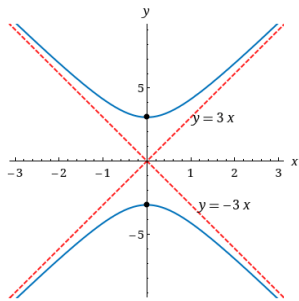
$$c^2 = a^2 + b^2$$

$$36 = 16 + b^2$$

$$b^2 = 20$$

$$\frac{x^2}{16} - \frac{y^2}{20} = 1$$

20) Find the standard equation for the hyperbola whose graph is shown



y-major

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$a = 3$$

$$\text{asymptote: } y = \pm \frac{a}{b} x$$

$$y = \pm 3x$$

$$\pm \frac{3}{b} x = \pm 3x$$

$$b = 1$$

$$\boxed{\frac{y^2}{9} - \frac{x^2}{1} = 1}$$

21) Consider a hyperbola with the following properties.

foci: $(\pm 10, 0)$, vertices: $(\pm 6, 0)$

Determine the values of a , b , and c .

$$\boxed{a = 6}$$

$$\boxed{c = 10}$$

$$c^2 = a^2 + b^2$$

$$100 = 36 + b^2$$

$$b^2 = 64$$

$$\boxed{b = 8}$$

vertices on x -axis so transverse axis is $\boxed{\text{horizontal}}$

Find standard equation that satisfies the conditions

$$\boxed{\frac{x^2}{36} - \frac{y^2}{64} = 1}$$

22) Find the standard equation for the hyperbola that satisfies the given conditions.

foci: $(0, \pm 4)$, vertices: $(0, \pm 1)$

y-major

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$a = 1$$

$$c = 4$$

$$c^2 = a^2 + b^2$$

$$16 = 1 + b^2$$

$$b^2 = 15$$

$$\boxed{\frac{y^2}{1} - \frac{x^2}{15} = 1}$$

23) Find the standard equation for the hyperbola that satisfies the given conditions.

Vertices: $(0, \pm 4)$, hyperbola passes through $(-5, 6)$

y-major

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$a = 4$$

$$\frac{36}{16} - \frac{25}{b^2} = 1$$

$$1 \times \frac{26}{4} = \frac{255}{b^2}$$

$$b^2=20$$

$$\frac{y^2}{16} - \frac{x^2}{20} = 1$$